Untitled3

September 9, 2026

[4]:
```
# Exercise 1: Display Prime Factors of a Number

def prime_factors(n):
    i = 2
    print("Prime factors are:")
    while i * i <= n:
        while n % i == 0:
            print(i, end=" ")
            n = n // i
        i = i + 1
    if n > 1:
        print(n)
num = int(input("Enter a number: "))
prime_factors(num)
```

```
Enter a number:  60
```

```
Prime factors are:
2 2 3 5
```

[5]:
```
# Exercise 2: Print All Even Numbers from 1 to 100 Using Function

def print_even():
    for i in range(1, 101):
        if i % 2 == 0:
            print(i, end=" ")
print("Even numbers from 1 to 100 are:")
print_even()
```

```
Even numbers from 1 to 100 are:
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56
58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96 98 100
```

[6]:
```
# Exercise 3: Quadratic Equations Roots with Exception Handling

import math

def quadratic_roots(a, b, c):
```

```
    d = b**2 - 4*a*c
    try:
        x = math.sqrt(d)
        if d == 0:
            r = -b / (2*a)
            print("Roots are real and equal")
            print("Root =", r)
        else:
            r1 = (-b + x) / (2*a)
            r2 = (-b - x) / (2*a)
            print("Roots are real and different")
            print("Root 1 =", r1)
            print("Root 2 =", r2)
    except ValueError:
        print("Error: math domain error")
a = float(input("Enter coefficient a:"))
b = float(input("Enter coefficient b:"))
c = float(input("Enter coefficient c:"))
quadratic_roots(a, b, c)
```

```
Enter coefficient a: 1
Enter coefficient b: -5
Enter coefficient c: 6

Roots are real and different
Root 1 = 3.0
Root 2 = 2.0
```

[ ]: